ALGEBRA 1 · UNIT 2: SOLVING LINEAR EQUATIONS

Food Truck Startup

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students run the numbers for a festival taco truck: they build cost, revenue, and profit expressions, solve equations for break-even and profit goals, and pitch a menu of their own design.

This project moves students from writing expressions to using equations as decision-making tools. Day 1 builds the model, Day 2 uses equations to answer business questions, and Day 3 is a design-and-pitch day where groups create their own menu item.

Standards Addressed

- A-SSE.1 Interpret parts of an expression in context
- A-CED.1 Create equations in one variable and use them to solve problems
- A-REI.3 Solve linear equations in one variable
- A-CED.4 Rearrange formulas to highlight a quantity of interest (Honors)

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the hero image and ask: How many tacos would you need to sell to make money? Collect guesses. Complete Part 1.
Day 2	Part 2: break-even and goal equations. Share solutions on the board and discuss rounding up in context.
Day 3	Part 3 design + pitch poster. Optional gallery walk where students vote on the best business plan.

Materials

- Student handout (one per student)
- Calculators
- Poster paper or slide template for the pitch
- Optional: real menu prices from a local food truck

How the Three Versions Differ

Level	What changes
CP2	One menu item, a single fixed cost ($120), a guided table, and sentence starters. Equations are two-step.
CP1	Two menu items, $150 in fixed costs, error analysis, solving for an unknown price, and comparing to a rival truck (variables on both sides).
Honors	Generalizes the business into the formula $P = (s - c)n - F$, rearranges literal equations, adds a 10% festival fee, and asks students to engineer a tie with the rival.

Watch For (Common Misconceptions)

- Adding costs instead of subtracting them when writing profit (watch the distributive property with −1).
- Reporting a decimal number of tacos. Students should round up to the next whole item to break even.
- Confusing revenue (money taken in) with profit (money kept).

Extension Ideas

- Have students research real ingredient costs and update the model.
- Graph the profit equation and connect the break-even point to the x-intercept (previews Unit 5).
- Add a sales tax or discount and discuss how the model changes.

Answer Key – CP2

Part 1: Build the Business Model

#	Task	Answer
1	Complete the table. Total cost = $120 booth fee + $1.50 for each taco. Revenue = $4.00 for each taco. Profit = revenue − total cost.	t = 10: cost 135, profit −95. t = 20: revenue 80, profit −70. t = 40: cost 180, revenue 160, profit −20. t = 60: cost 210, revenue 240, profit 30.
2	Write an expression for the total cost of making t tacos.	120 + 1.50t
3	Write an expression for the revenue from selling t tacos.	4t
4	Profit = revenue − cost = 4t − (120 + 1.50t). Simplify this expression. Show each step.	4t − 120 − 1.50t = 2.50t − 120
5	Use your profit expression to find the profit if you sell 100 tacos.	2.50(100) − 120 = $130

Part 2: Answer the Council's Questions

#	Task	Answer
6	Break-even means profit = $0. Write and solve an equation to find how many tacos you must sell to break even.	2.50t − 120 = 0 → 2.50t = 120 → t = 48 tacos
7	Your group wants to earn $300 in profit. How many tacos must you sell?	2.50t − 120 = 300 → 2.50t = 420 → t = 168 tacos
8	You add lemonade: it costs $0.50 to make and sells for $2.00. If you sell 40 tacos, how many cups of lemonade must you sell to reach a profit of $100?	2.50(40) − 120 + 1.50L = 100 → −20 + 1.50L = 100 → 1.50L = 120 → L = 80 cups
9	What if you raise the taco price to $4.50? Write a new profit expression and find the new break-even number of tacos. How many fewer tacos is that?	Profit = 3t − 120. 3t − 120 = 0 → t = 40 tacos, which is 8 fewer than 48.

Part 3: Design & Pitch

#	Task	Answer
10	Create a new menu item. Choose its name, its cost to make, and its selling price.	Answers vary. Profit per item = selling price − cost to make.
11	If your new item had to pay the $120 booth fee by itself, how many would you need to sell to break even? Write and solve an equation.	Answers vary: (profit per item) · n − 120 = 0. Round up to the next whole item.

12	Write 2–3 sentences recommending a plan to the town council. Use at least two numbers from your work.	Answers vary; must cite numbers from the work.

Answer Key – CP1

Part 1: Build the Business Model

#	Task	Answer
1	Write an expression for the total cost C of making t tacos and b burritos (include all fixed costs).	C = 150 + 1.50t + 2.50b
2	Write an expression for the revenue R from selling t tacos and b burritos.	R = 4t + 7b
3	Write and simplify an expression for the profit, R − C.	P = 4t + 7b − (150 + 1.50t + 2.50b) = 2.50t + 4.50b − 150
4	Evaluate your profit expression for 30 tacos and 20 burritos. What does the answer tell you?	2.50(30) + 4.50(20) − 150 = 75 + 90 − 150 = $15; the truck makes only $15 profit.
5	Jordan wrote the profit as 4t + 7b − 150 + 1.50t + 2.50b = 5.50t + 9.50b − 150. Explain Jordan's mistake and how to fix it.	Jordan added the costs. The whole cost must be subtracted: −(150 + 1.50t + 2.50b), so P = 2.50t + 4.50b − 150.

Part 2: Answer the Council's Questions

#	Task	Answer
6	If you sold only tacos, how many would you need to sell to break even?	2.50t − 150 = 0 → t = 60 tacos
7	At past festivals, trucks sold twice as many tacos as burritos (t = 2b). Rewrite the profit using only b. Then find how many of each item you must…	2.50(2b) + 4.50b − 150 = 9.50b − 150. 9.50b − 150 = 420 → 9.50b = 570 → b = 60 burritos, t = 120 tacos
8	Suppose you sell 120 tacos and 60 burritos. What burrito price p would give a profit of exactly $480? (Tacos stay at $4.00.)	2.50(120) + 60(p − 2.50) − 150 = 480 → 300 + 60p − 150 − 150 = 480 → 60p = 480 → p = $8.00

Part 3: Face the Competition & Pitch

#	Task	Answer
9	Write a profit expression for Burger Barn and one for your truck if you sold only tacos. Write and solve an equation to find when the two trucks…	Burger Barn: 2t − 90. Ours: 2.50t − 150. 2.50t − 150 = 2t − 90 → 0.50t = 60 → t = 120 items (both earn $150).
10	Graph both profit expressions on the grid. Who is ahead before the tie point? After?	Lines cross at (120, 150). Burger Barn is ahead for fewer than 120 items; your truck is ahead after 120.
11	Add a third menu item of your own design. Write the new profit expression and find two different combinations of items that would earn $500.	Answers vary; check that the new term is (price − cost) times the variable and that both combinations total $500.

Answer Key – Honors

Part 1: Generalize the Model

#	Task	Answer
1	Write a formula for the profit P from selling n items of one kind. Show that it can be written as P = (s − c)n − F.	P = sn − cn − F = (s − c)n − F
2	Solve the formula for n.	n = (P + F) ÷ (s − c)
3	Use your formula to find how many tacos must be sold for a profit of $600 (F = $150).	n = (600 + 150) ÷ (4 − 1.50) = 750 ÷ 2.50 = 300 tacos
4	Explain why s − c must be positive for the business to work. What happens to your formula for n if s = c?	If s − c ≤ 0, each sale loses money (or makes none), so profit can never cover F. If s = c the formula divides by 0, so there is no solution.

Part 2: Add the Festival Fee

#	Task	Answer
5	Write and simplify a profit expression for selling t tacos and b burritos after the 10% fee.	0.90(4t + 7b) − (150 + 1.50t + 2.50b) = 3.60t + 6.30b − 150 − 1.50t − 2.50b = 2.10t + 3.80b − 150
6	If t = 2b, rewrite the profit using only b and find how many of each item you must sell to make $650.	2.10(2b) + 3.80b − 150 = 8b − 150. 8b − 150 = 650 → 8b = 800 → b = 100 burritos, t = 200 tacos
7	For a burrito-only truck with the fee, P = (0.90s − c)n − F. Solve this formula for s. Then find the burrito price needed to make $500 by selling…	P + F = (0.90s − c)n → (P + F) ÷ n + c = 0.90s → s = (P + F + cn) ÷ (0.90n). s = (500 + 150 + 250) ÷ 90 = 900 ÷ 90 = $10.00

Part 3: Engineer the Competition

#	Task	Answer
8	Find the number of items at which your taco-only profit equals Burger Barn's profit.	2.50t − 150 = 2t − 90 → 0.50t = 60 → t = 120 items
9	The council will lower your fixed cost if it helps. What fixed cost F would make the two trucks tie at exactly 80 items? Show your equation.	2.50(80) − F = 2(80) − 90 → 200 − F = 70 → F = $130
10	Graph your original profit line, your new profit line (with the new F), and Burger Barn's line. Explain what changing F does to the graph.	Lowering F from 150 to 130 shifts the line up 20 units (same slope), moving the tie point from 120 to 80 items.
11	Design a two-item menu of your own (with the 10% fee). Write a general profit formula, choose prices, and show that your plan earns at least $500…	Answers vary; formula should have the form $(0.90s_1 - c_1)x + (0.90s_2 - c_2)y - F$.

Scoring note: Accept equivalent forms of rearranged formulas, e.g., $n = (P + F) \div (s - c)$ or $n = P \div (s - c) + F \div (s - c)$.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original menu item has a realistic cost and price with a correct profit equation. Goes beyond the minimum with a creative, realistic choice.	Original menu item has a realistic cost and price with a correct profit equation.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Pitch poster shows the menu, profit model, break-even point, and a recommendation. Polished, organized, and visually engaging.	Pitch poster shows the menu, profit model, break-even point, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.